Journal Supplementary G — Limitations and threats to validity

Companion to: *Beyond Accuracy* (Section VII)

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A.1 Scope

This supplement enumerates the limitations and threats to validity of the twelve-axis framework at greater depth than journal Section VII admits. We organise them as (a) design-time constraints baked into the framework, (b) empirical caveats specific to the present benchmark set, (c) statistical caveats around the inference, and (d) reporting choices that may need revision in follow-up work.

B.2 Design-time constraints

B.2.1 Canonical K is held fixed within an axis

Every axis is computed at the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ except for F-4 (capacity sweep). This is by design: changing K between axes would introduce a second degree of freedom that mixes with the axis-specific perturbation (seed / band / preprocessing / scene). The trade-off is that the per-axis numbers depend on a non-trivially chosen K that a future paper may want to vary; the framework admits per-axis K-sweeps as a forward-compatible extension without redesign.

B.2.2 Uniform tokenisation across scenes

$Q + 1 = 13$ quantisation levels is fixed across scenes. Scenes with narrow spectral dynamic range (Pavia U, ROSIS, 8-bit per band) and wide dynamic range (AVIRIS, 12-bit) are tokenised at the same resolution, which may waste resolution on the narrow-range scenes. A future paper exploring per-scene adaptive Q should re-run all axes; the framework will automatically pick up the new Q from the canonical-fit hyperparameter slate.

B.2.3 Hyperparameters (α, η) held fixed

$\alpha = 0.45, \eta = 0.20$ across all fits. These are scikit-learn LDA defaults that we have not tuned per scene. The companion code repository runs a hyperparameter grid in `lda_hyperparam_search/` for reference but the framework's canonical fit deliberately fixes the slate to keep cross-axis comparisons clean.

B.2.4 Stratified 220-per-class sampling

The canonical fit samples 220 pixels per admissible class without replacement. This caps the per-class effective budget regardless of how many labelled pixels a class actually has. Classes with very few labelled pixels (alfalfa on Indian Pines, oats) are dropped from the canonical fit on the standard ≥ 100-pixel admissibility test. The framework reports the admissibility decisions in the per-scene JSON metadata.

C.3 Empirical caveats specific to the benchmark set

C.3.1 Indian Pines is small and class-imbalanced

At 145×145 with 16 classes and a long tail (some classes have < 100 labelled pixels), Indian Pines is the smallest of the six labelled scenes. The per-axis numbers on Indian Pines should be treated with proportionate caution.

C.3.2 Salinas-A is a subset of Salinas

Salinas-A's six classes are a strict subset of Salinas's sixteen, and the spatial footprint of Salinas-A is a contiguous rectangle within Salinas. Numbers across these two "scenes" are *not* independent. A future paper may report only one of the two; we report both because the band-mask diagnostic surfaces a qualitatively different finding on Salinas-A (SWIR-only paired ARI = 0.766) than on Salinas (SWIR-only paired ARI = 0.435), and the contrast is itself a finding.

C.3.3 Pavia U is VNIR-only by sensor design

The ROSIS-03 sensor on Pavia U covers 430–860 nm, which is entirely VNIR. The SWIR mask consequently retains 0 bands and is auto-skipped. The no-water mask is a no-op on Pavia U (no bands fall in the water-vapour ranges) and produces the same paired ARI as the VNIR mask. These artefacts are recorded transparently in the per-(scene, mask) JSON outputs.

C.3.4 Botswana and KSC have wetland-dominant land cover

The two scenes that consistently appear in the band-fragile regime share a wetland-dominant land cover. Whether band-fragility is a property of the LDA fit on these scenes, of the canonical hyper-parameter slate, or of the wetland-spectral-signature itself is not resolvable from the present data; we flag the open question.

D.4 Statistical caveats

D.4.1 Hierarchical Bayesian model assumes Gaussian residuals

The F-1 likelihood $\epsilon_{m,s,f} \sim \mathcal{N}(0, \sigma^2)$ is a standard working assumption for OA-scaled residuals on $[0, 1]$. On the present benchmarks the residuals are not visibly non-Gaussian (no heavy tails, no bimodality observed in the posterior predictive) but a Student-t alternative may be appropriate on a benchmark set with a strong outlier.

D.4.2 Pairwise $P[A > B]$ multiplicity correction

On 5 methods there are $5 \cdot 4 = 20$ ordered pairs; without correction the family-wise error rate inflates with the number of implicit comparisons in the reader's interpretation. We adopt a **Bayesian-FDR-style** adjustment: the reported posterior $P[A > B]$ already factors out scene + fold random effects in the hierarchical model, so the multiplicity correction reduces to discounting the small marginal posterior tails. Bonferroni at $\alpha = 0.05$ on 20 ordered pairs requires a per-pair threshold of $P > 0.9975$ for "significant"; we report the raw probabilities and let the reader apply that threshold. Our headline finding $P[\text{topic\_routed\_soft} > \text{raw\_logistic}] = 0.641$ is *not* significant under either Bonferroni $\alpha = 0.05$ or an uncorrected $\alpha = 0.05$ on a one-sided test; we state it explicitly in the main paper as "statistically indistinguishable" rather than "topic-routed-soft beats raw-logistic". The HIDSAG inversion $P[\text{topic\_logistic} > \text{raw\_logistic}] = 0.0$ *is* significant under any reasonable correction.

D.4.3 Seed-stability $N = 5$ is small for multi-modal posteriors

The within-method standard deviation across 5 seeds is below 0.03 on all reported cells, suggesting a unimodal posterior on the present benchmarks. KSC and Botswana, on which the canonical basis appears weakly identified, may have multi-modal posteriors that the 5-seed budget would not reveal. The companion code repository's `deep_seed_stability/` block runs $N = 30$ seeds; the framework's $N = 5$ choice is for parsimony with an explicit upgrade path.

D.4.4 c_v co-occurrence statistics are computed on the same corpus

The c_v reference corpus for the co-occurrence count is the canonical document-term matrix of the same scene. Computing c_v on an external reference corpus (a published HSI library, for example) would yield a different number and is a worthwhile follow-up.

E.5 Reporting choices

E.5.1 Per-axis JSON over a single combined CSV

The framework ships one JSON per axis per scene rather than a single combined CSV. This is by design (clean separation of provenance) but creates a slight friction for follow-up analysis that wants to join across axes. The companion code repository provides a `manifests/` index that lists every artefact path and its schema, intended for join-on-load workflows.

E.5.2 Figure choices in the main paper

The journal main paper carries 4 figures (paired ARI heatmap, Hungarian alignment, Bayesian forest plot, basis spectra grid). The supplement carries the remaining 5 (coherence-vs-ARI, seed-stability swarm, capacity sweep, cross-method 8x8 grid, HIDSAG band-mask). The split reflects journal-section relevance, not figure quality ordering.

E.5.3 No statistical significance test for paired ARI

Paired ARI does not carry a built-in significance threshold. We do not compute permutation-test p-values per (scene, mask) tuple because the test is computationally expensive on $D \sim 3000$ pixels and the framework's per-(scene, mask) values are already extreme (paired ARI ≈ 0.01 on KSC vs 0.77 on Salinas-A SWIR). A future paper applying the framework on a marginal-effect benchmark would benefit from a permutation companion.

F.6 What we would change in a 2nd-paper redesign

If we were to redesign the framework from scratch given everything we now know:

1. Move per-axis K-sweep from a separate axis (F-4) to a layer that wraps every other axis: report (F-1, F-2, ..., F-12) at $K \in \{4, 6, 8, 10, 12, 16\}$ rather than at the canonical K only.

2. Replace F-12 (external baseline OA delta) with an internal-comparison axis: compare the canonical topic-routed-soft against a fixed deep baseline trained inside the framework on the same train/test splits. The current external baseline carries the cross-paper-comparison caveat that the deltas are not interpretable without the per-paper split.

3. Add a per-pixel-confidence axis: report $\Pr[\text{argmax}_k \theta_d = z_d^* \mid \theta_d]$ calibrated against a held-out validation fold. This would expose the calibration of the topic representation directly.

None of these changes invalidate the present framework; they identify directions for forward-compatible extensions.